5 **(a)** Fig. 5.1 shows the variation with voltage V of the current I across a filament lamp rated 6.0 V, 1.5 W.

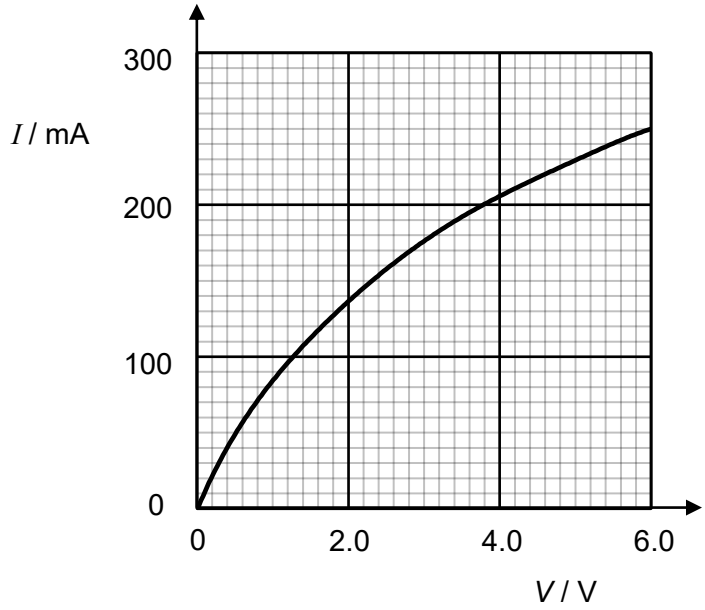

Fig. 5.1

(i) In microscopic terms, explain why the resistance of the filament lamp increases as V increases.

.....

.....

.....

.....

.....

.....

.....[3]

(ii) On Fig. 5.1, draw a line to show the variation with V of I if the filament lamp is assumed to be ohmic. [1]

(iii) Use Fig. 5.1 to determine the change in resistance of the filament lamp when V increases from 4.0 V to 6.0 V.

change in resistance = ... Ω [2]

(b) A student designs a circuit for a night light using the filament lamp in **(a)** and a light-dependent resistor (LDR), as shown in Fig. 5.2.

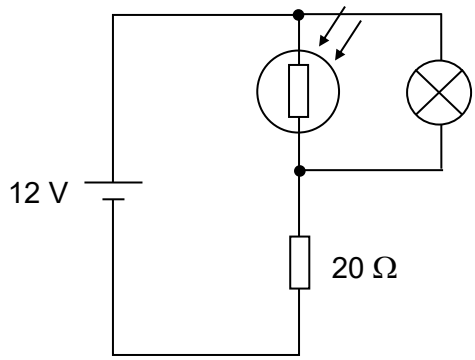

Fig. 5.2

The LDR has a resistance of 10 Ω in daylight and increases to 1000 Ω in the dark.

(i) Explain why the lamp will not operate at its rated power in daylight.

…….....

…….…….... [2]

(ii) Calculate the resistance of the LDR in order for the lamp to operate at its rated power of 1.5 W.

resistance = ……. Ω [3]